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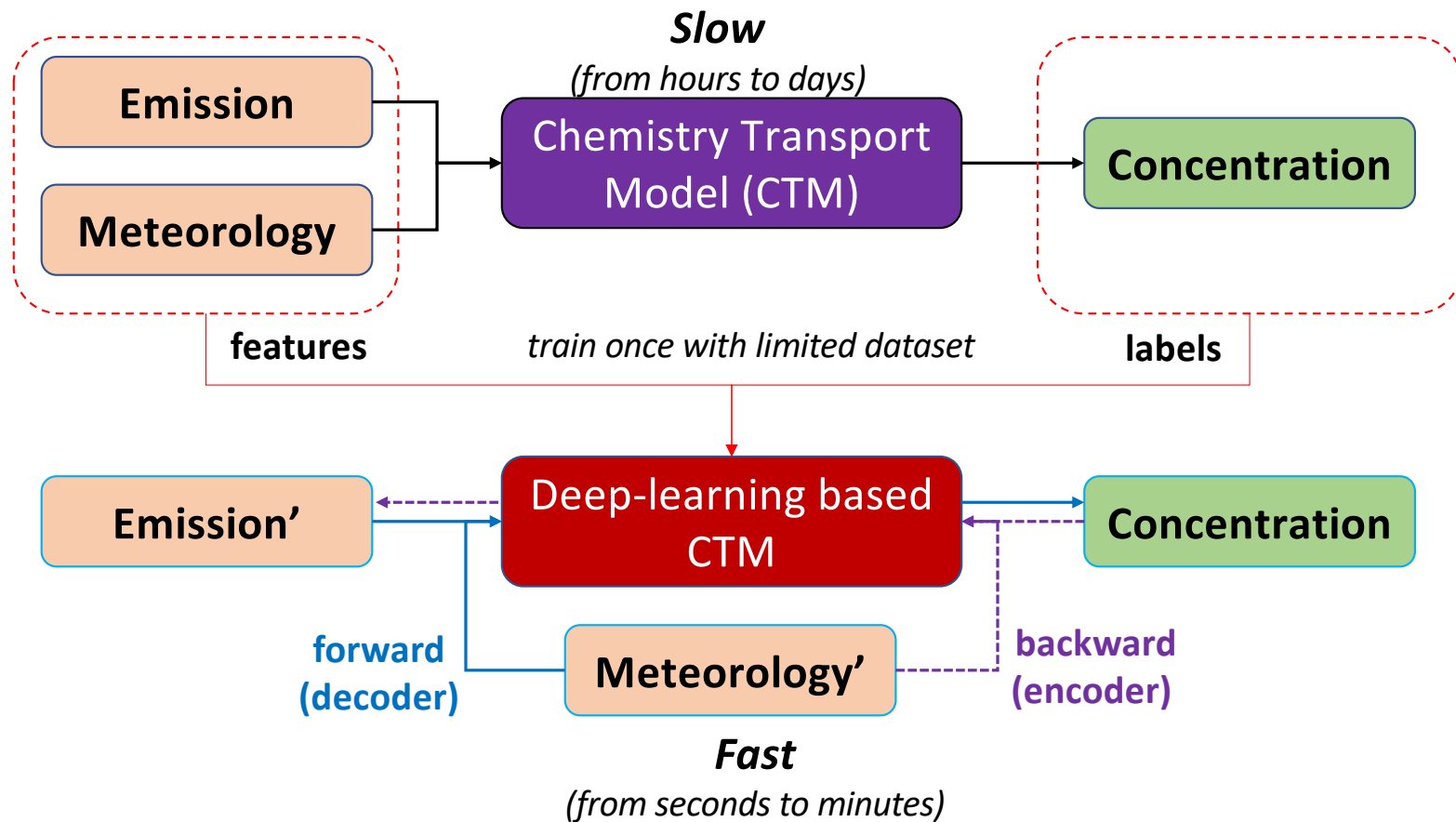
Deep-learning-based Chemical Transport Model and Inverse Model Applications and Uncertainties

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September 17, 2026

Main Objectives



DeepCTM: Mimic the CTM modeling

Input1: Emissions

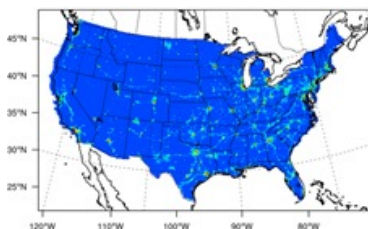
(Row, Col, Ht, Time)

5 key pollutants:

- SO_2 , NO_x , NH_3 , VOC, PM

6 key sectors:

- Power
- Industry
- Domestic
- Transport
- Agriculture

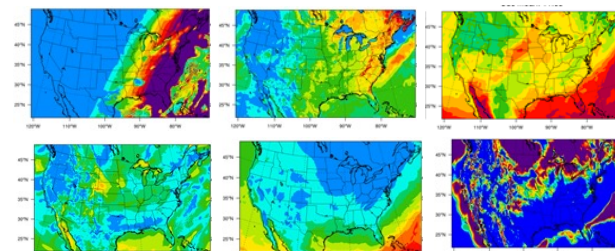


Input2: Meteorology

(Row, Col, Ht, Time)

selected key variables:

- T2, Q2, WS, WD, PBL
- Radiation, Precipitation, etc

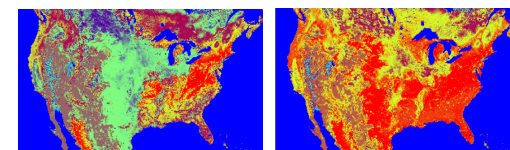


Input3: Geography

(Row, Col)

selected key variables:

- LAI, land cover, elevation, etc



Output: Concentrations

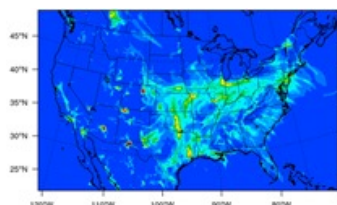
(Row, Col, Ht, Time)

4 key pollutants:

- SO_2 , NO_2 , O_3 , $\text{PM}_{2.5}$

Selected other species:

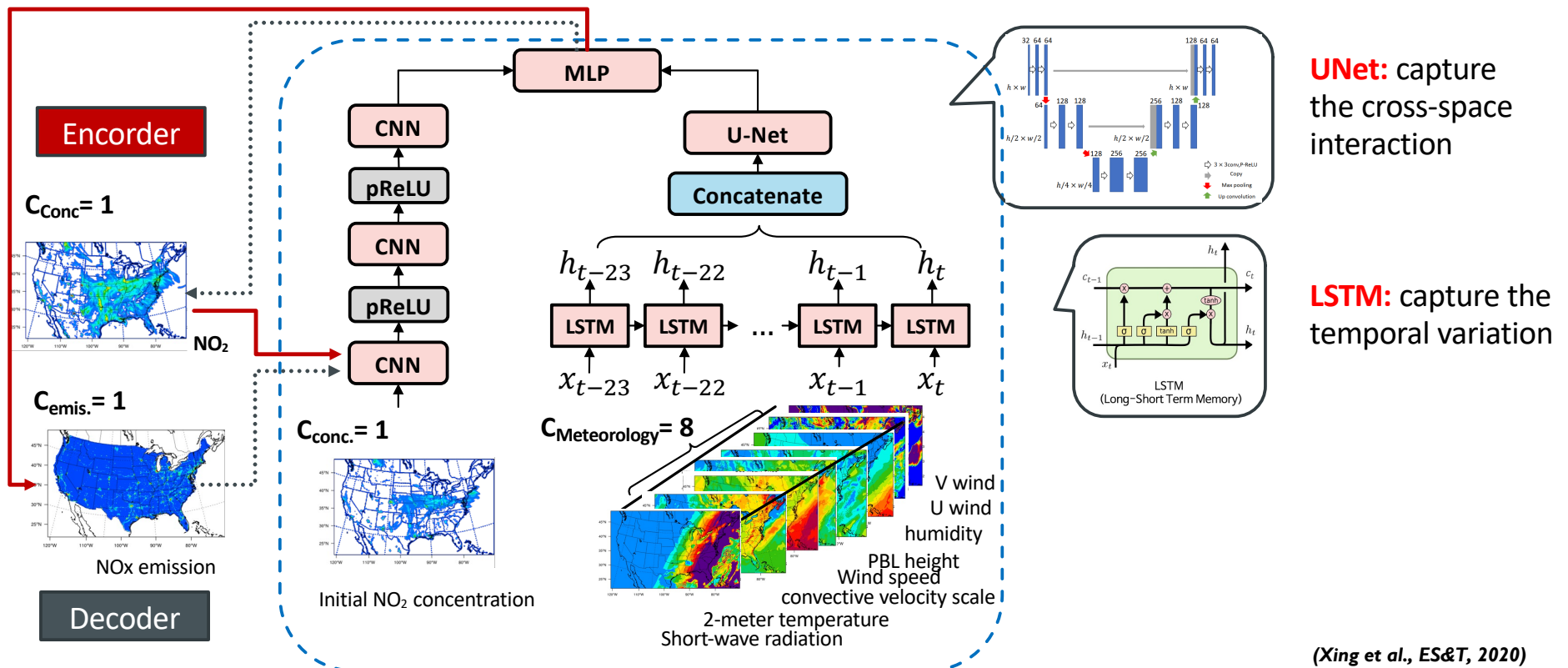
- NH_3 , VOC, PM species (sulfate, nitrate, SOA)



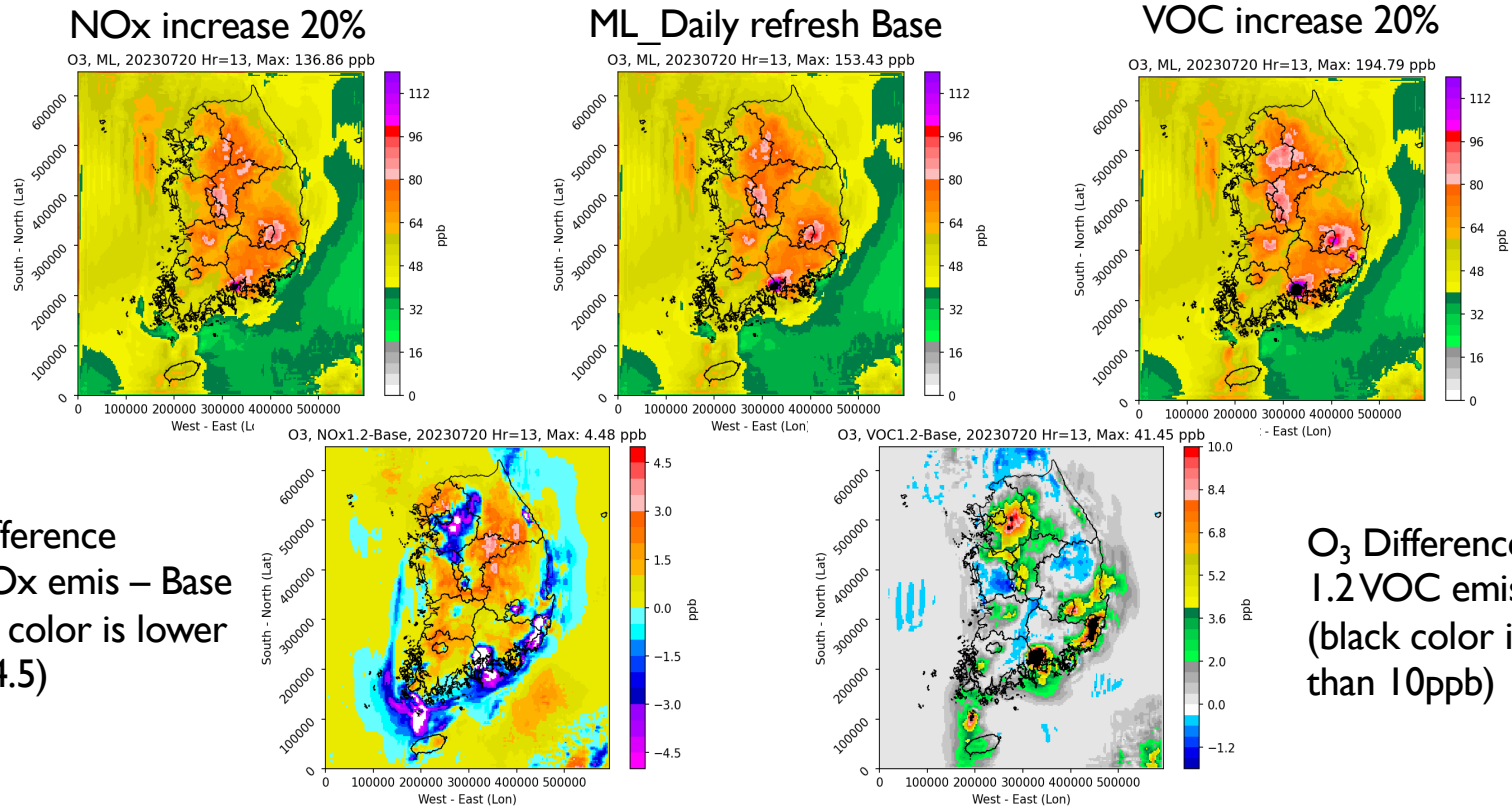
DeepCTM

- Deterministic system
- Mathematical equations based on physical laws
- Influenced by neighbor grids and previous days

DeepCTM (Decoder) & Inverse (Encoder) Modeling System



DeepCTM-O₃ Sensitivity Simulations [MDA8 O₃ on July 29, 2025]



In VOC limited region, NOx increase cause reduction of O₃, but NOx limit area can increase O₃

In VOC limited region, VOC increase cause O₃ increase

Annual Mean FNR (HCHO to NO₂ ratio) and O₃

2008

2011

2012

2014

2019

2020

FNR,w_unconv.2008,Summer_Average (JJAS)

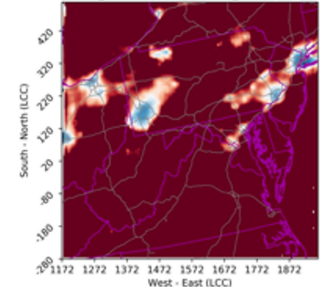
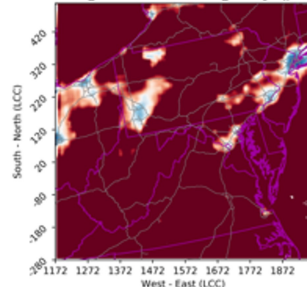
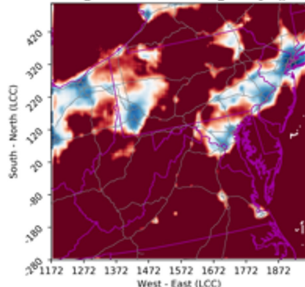
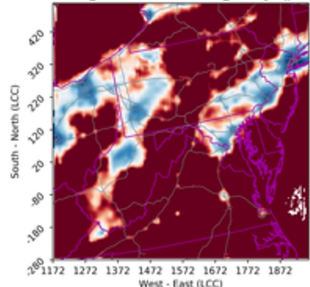
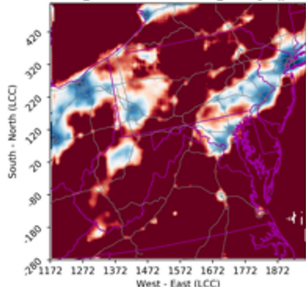
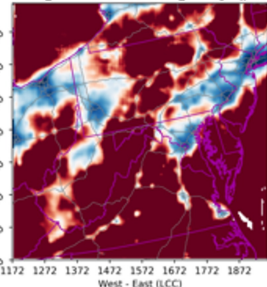
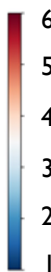
FNR,w_unconv.2011,Summer_Average (JJAS)

FNR,w_unconv.2012,Summer_Average (JJAS)

FNR,w_unconv.2014,Summer_Average (JJAS)

FNR,w_unconv.2019,Summer_Average (JJAS)

FNR,w_unconv.2020,Summer_Average (JJAS)



Delta FNR

FNR,Diff (w_unconv - wo_unconv), 2009 Summer_Average (JJAS)

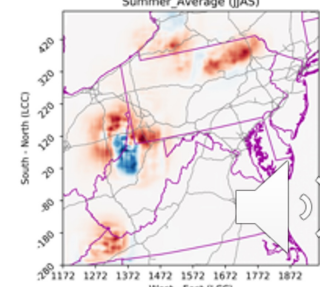
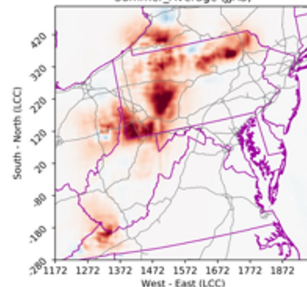
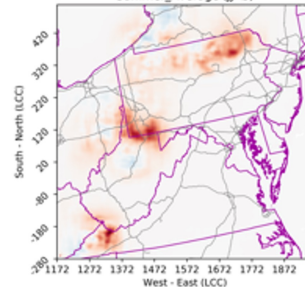
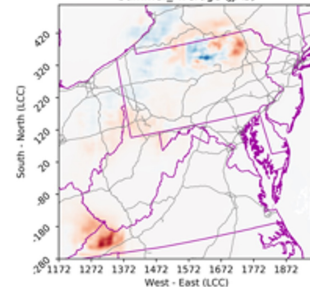
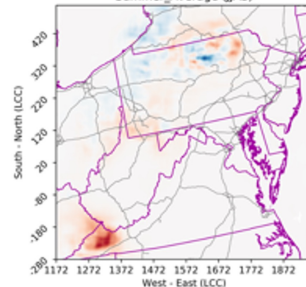
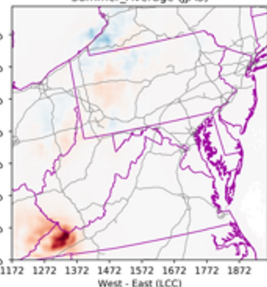
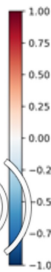
FNR,Diff (w_unconv - wo_unconv), 2011 Summer_Average (JJAS)

FNR,Diff (w_unconv - wo_unconv), 2012 Summer_Average (JJAS)

FNR,Diff (w_unconv - wo_unconv), 2014 Summer_Average (JJAS)

FNR,Diff (w_unconv - wo_unconv), 2019 Summer_Average (JJAS)

FNR,Diff (w_unconv - wo_unconv), 2020 Summer_Average (JJAS)



Delta O₃

O3,Diff (w_unconv - wo_unconv) Summer_Average (JJAS) Max: 0.57 ppb Min: -2.23

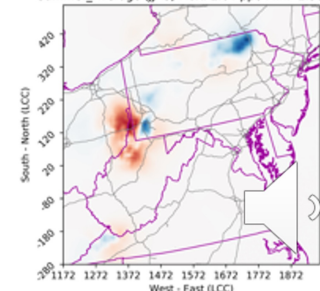
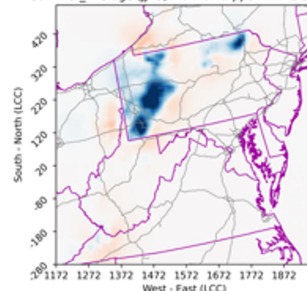
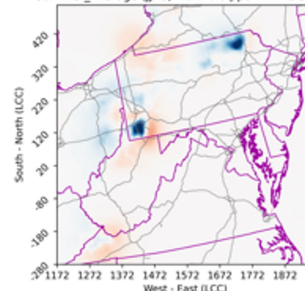
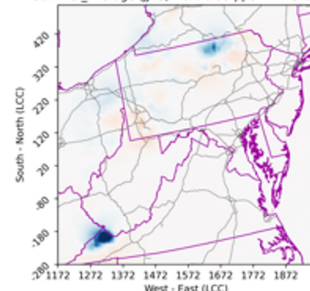
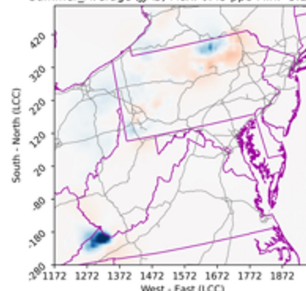
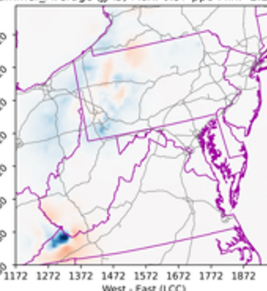
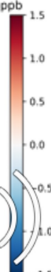
O3,Diff (w_unconv - wo_unconv) Summer_Average (JJAS) Max: 0.43 ppb Min: -3.15

O3,Diff (w_unconv - wo_unconv) Summer_Average (JJAS) Max: 0.33 ppb Min: -3.12

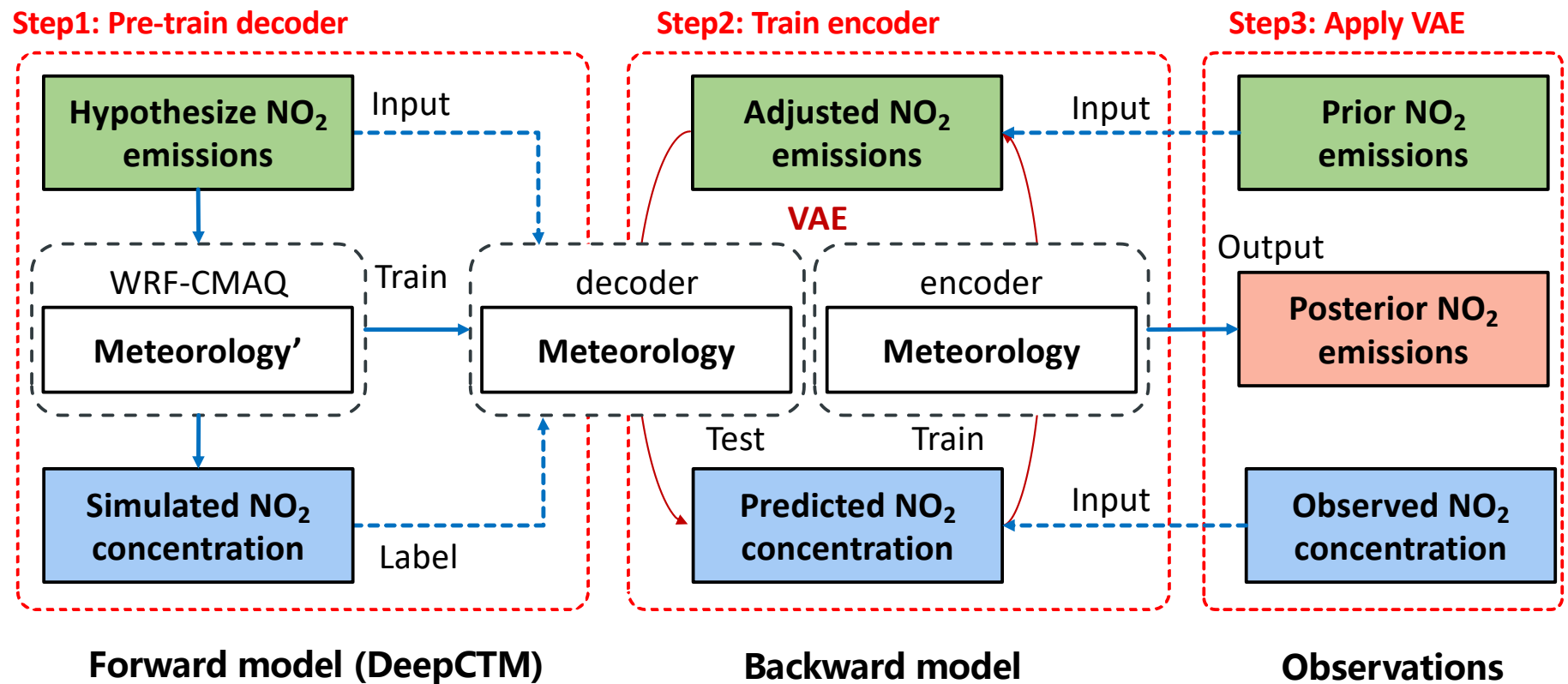
O3,Diff (w_unconv - wo_unconv) Summer_Average (JJAS) Max: 0.6 ppb Min: -2.35

O3,Diff (w_unconv - wo_unconv) Summer_Average (JJAS) Max: 0.41 ppb Min: -2.32

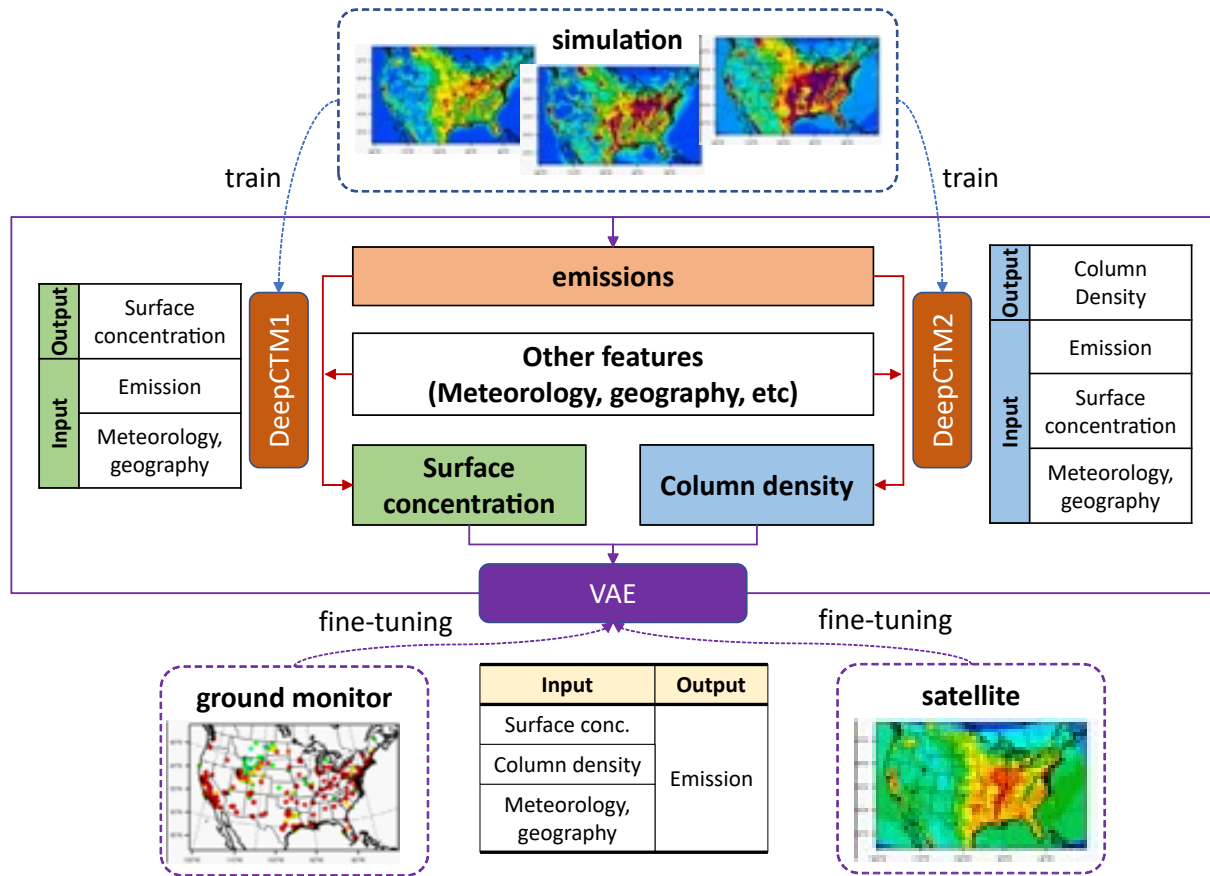
O3,Diff (w_unconv - wo_unconv) Summer_Average (JJAS) Max: 1.64 ppb Min: -1.38 ppb



Variational AutoEncoder (VAE) Model Structure

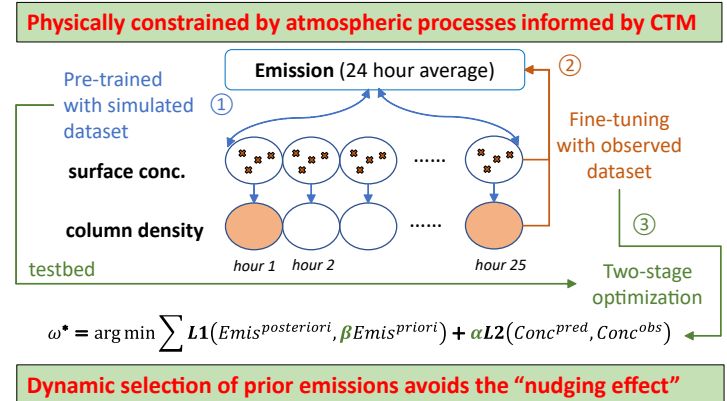


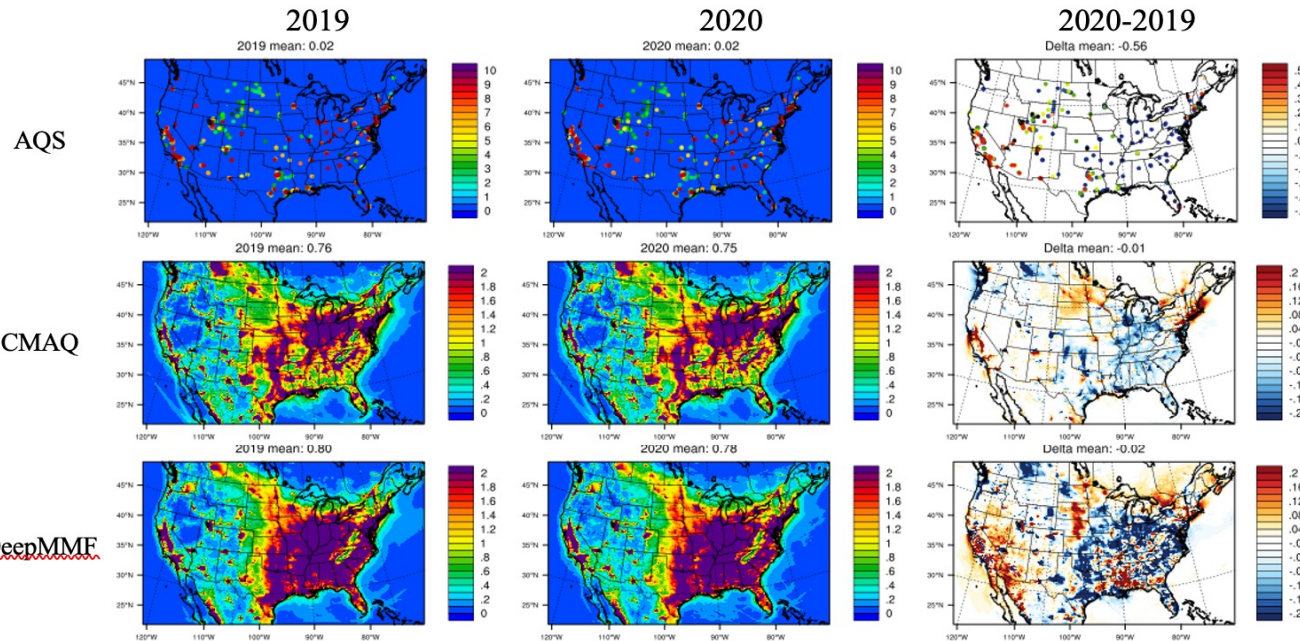
DeepMMF Inverse Modeling System



DeepMMF =
Deep-learning Model-Measurement Fusion

A combination of VAE inverse model with
observations tuning (fusion)

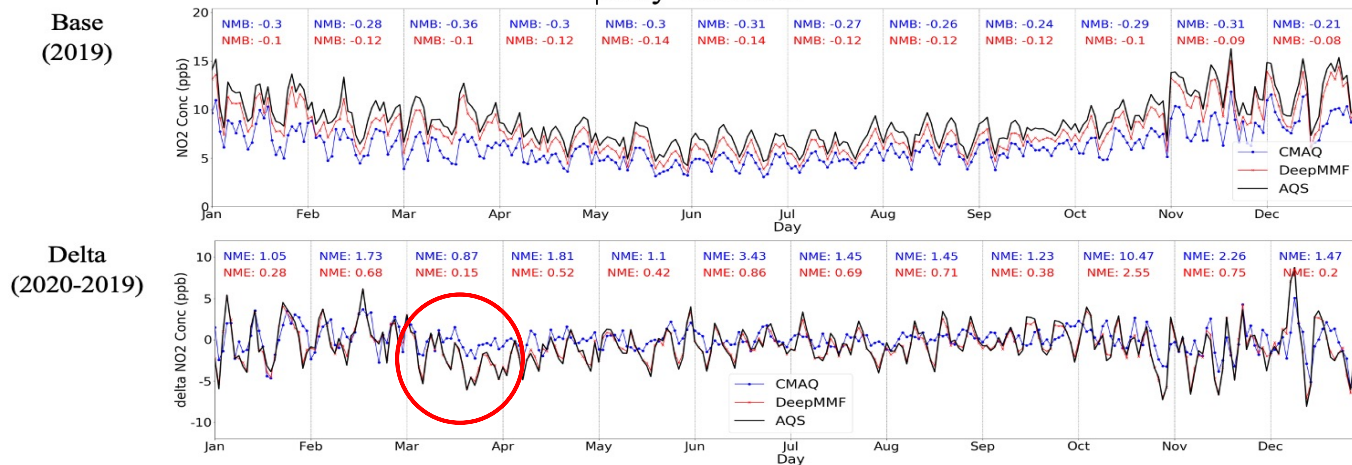




Surface DeepMMF

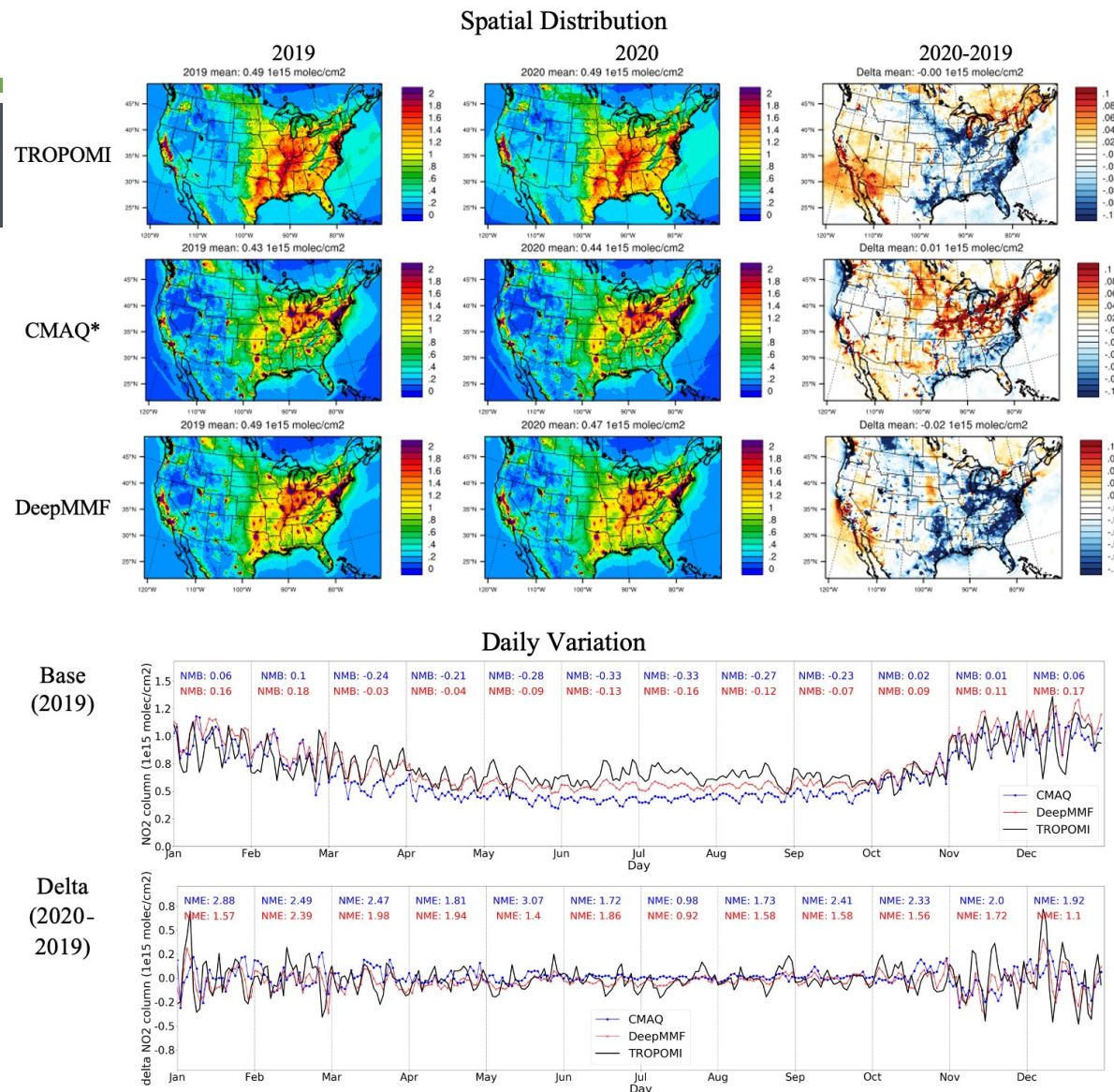
- CMAQ and DeepMMF show similar spatial patterns
- DeepMMF fused with AQS shows higher than CMAQ
- Suggests that CMAQ underestimates NO₂
- Same emissions with different met. for 2020
CMAQ fused with AQS → Reflects reductions of NO₂
- Lower impacts of met.

Daily Variation



Column Density DeepMMF

- Similar to AQS fusion
- Higher NO₂ column density from DeepMMF compared to CMAQ due to the higher density NO₂ from TROPOMI
- 2020 CMAQ = Same Emissions with 2020 Meteorology
- 2020 DeepMMF reflects the changes in TROPOMI NO₂ changes between the years
- Similar spatial patterns of the decrease of NO₂ in the Eastern U.S. vs the increase in the Western U.S.

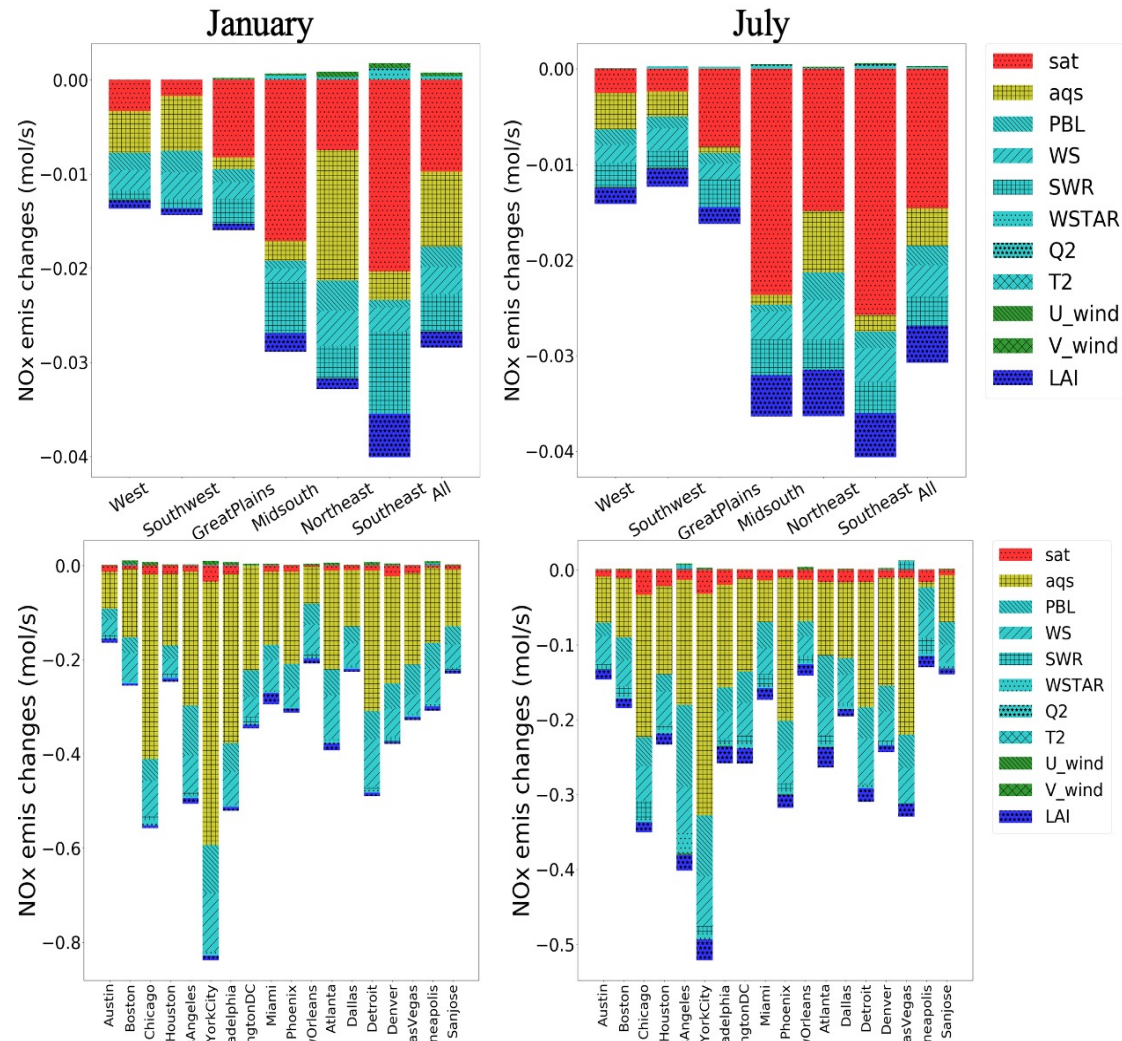


Source Attributions

Region

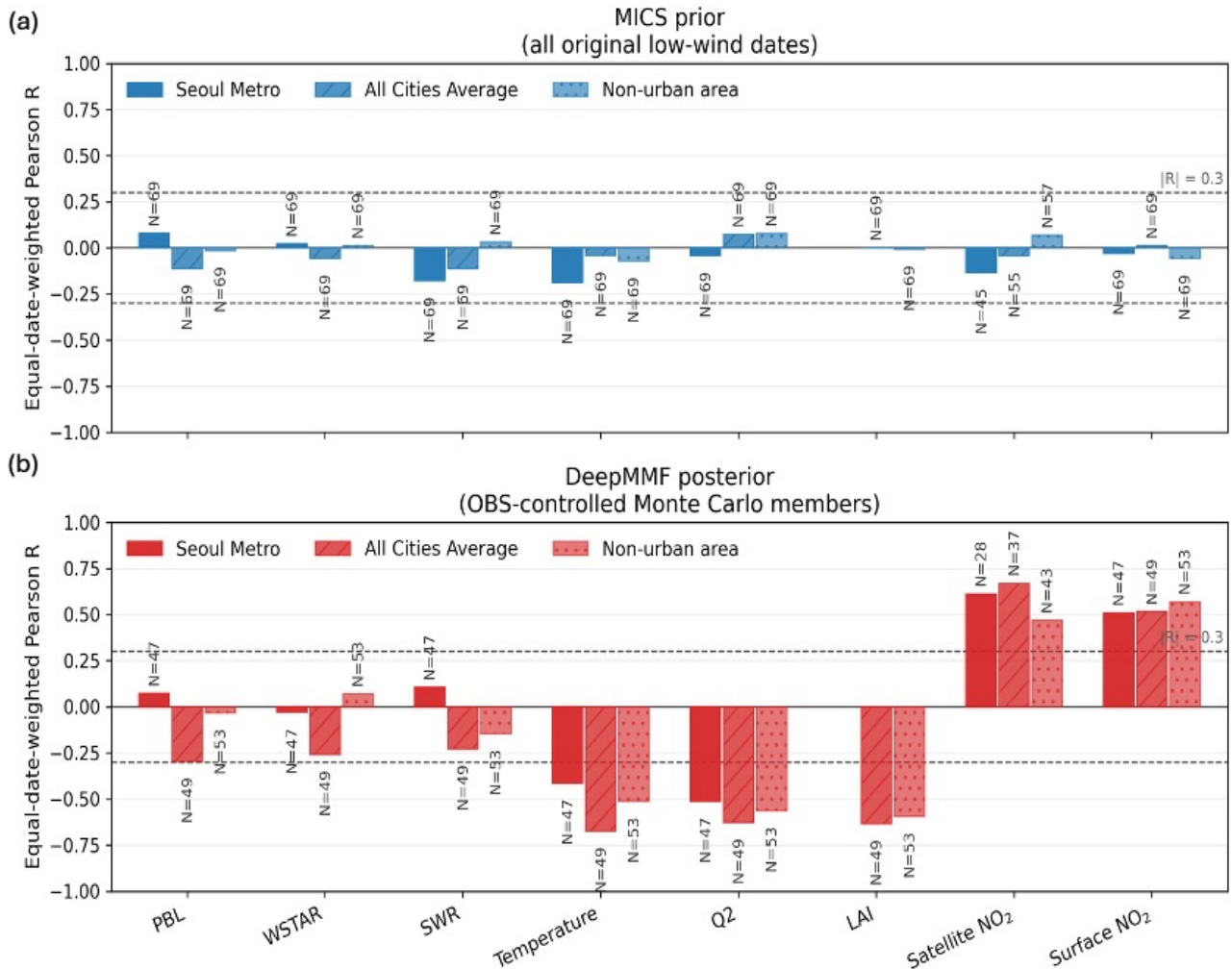
- Observations contribute to NO_x changes (2019-2020) > 50%
- Lower conc. → Lower emissions
- Met variables influence the NO_x reduction (e.g., PBL, WS, Short-Wave-Radiation)
- Lower PBL, WS, SWR → Stable dispersion with lower SWR → NO₂ reduction/oxidation
- Regional LAI changes → Bigger NO_x Reduction than the city
- **Regional VAE emissions are driven by Satellites**
- **City-level VAE emissions are driven by AQS**

City



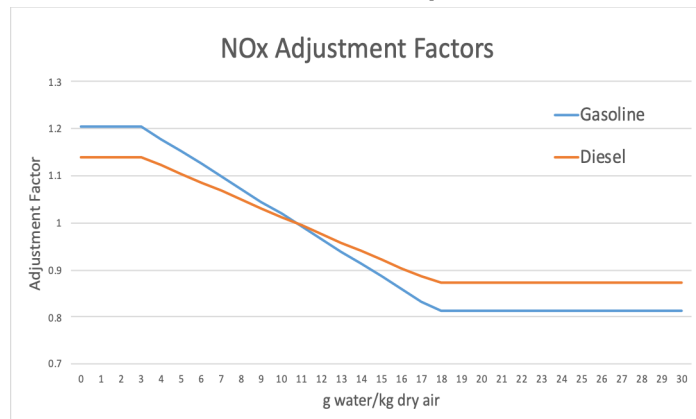
Sensitivity Study

- Focusing on Low windy weekdays only
- Clear correlations with T2, Q2, LAI, and the surface and satellite NO₂ observations
- Over the Seoul metro, no correlation with LAI due to the low density of green area over Seoul compared to other cities



NOx-Met Dependency

Mobile NOx humidity corrections



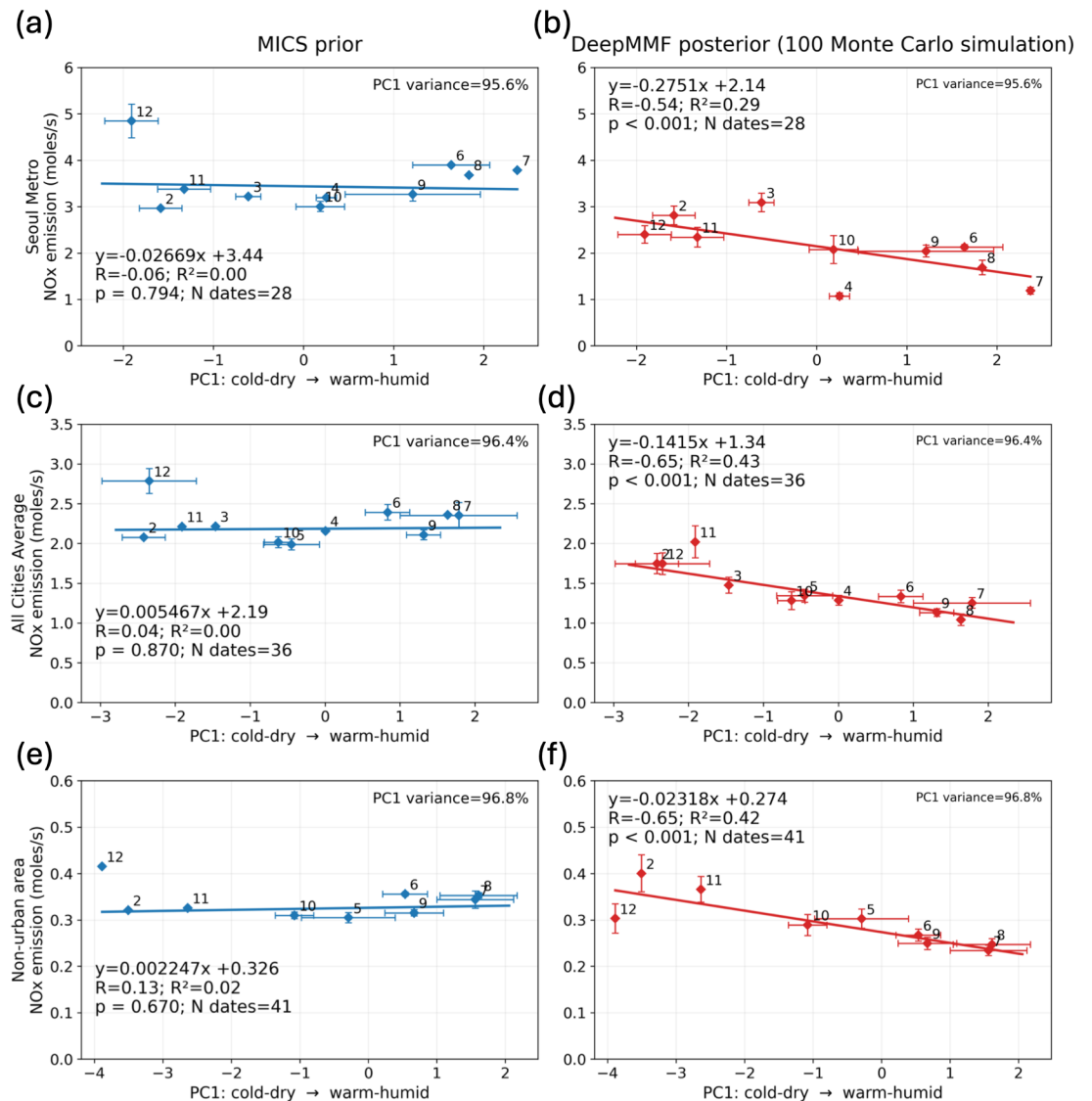
Residential heating NOx higher during the drier winter

Principal Component Analysis (PCA)

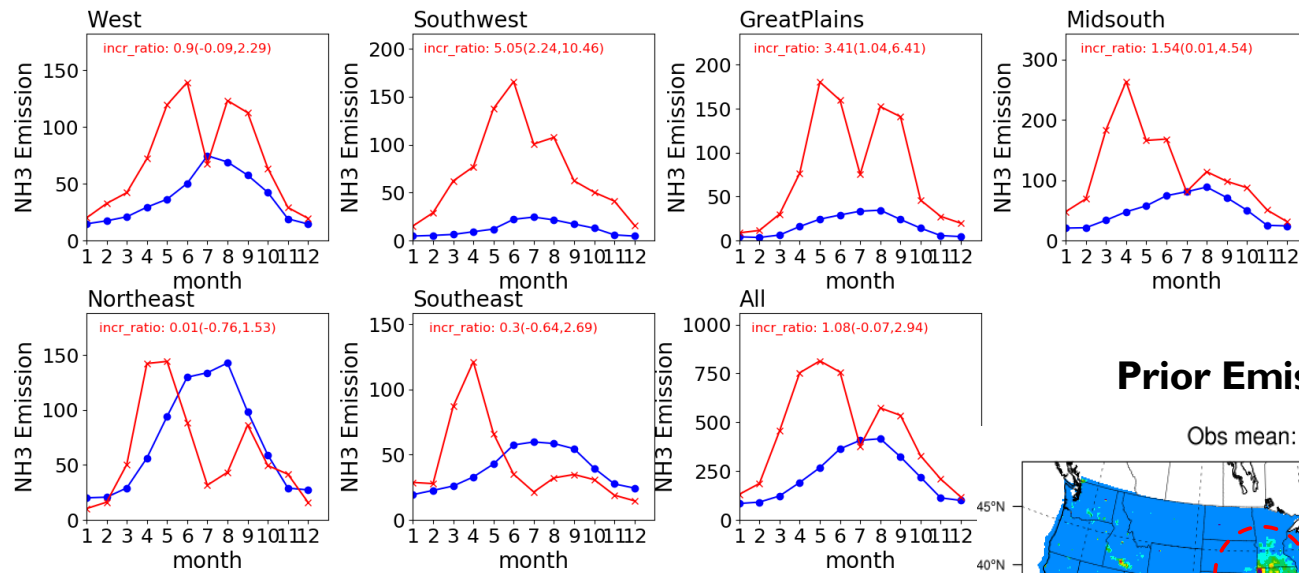
$$PC1 = 0.707T_2 + 0.707Q_2$$

$$PC2 = -0.707T_2 + 0.707Q_2$$

- NOx emissions shows the clear negative correlation with humidity and temperature



Monthly BU-TD NH₃ Emissions Comparison

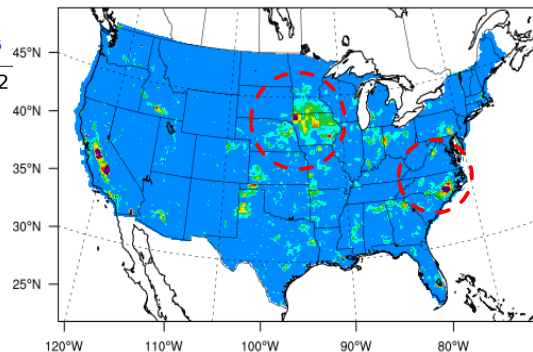


- Indicating that the significant differences over the Great Plains and Southwest regions

- Important to utilize the spatial and temporal coverages from the Crls NH₃
- **Relatively** underestimated NH₃ from Beef vs overestimated NH₃ from Swine

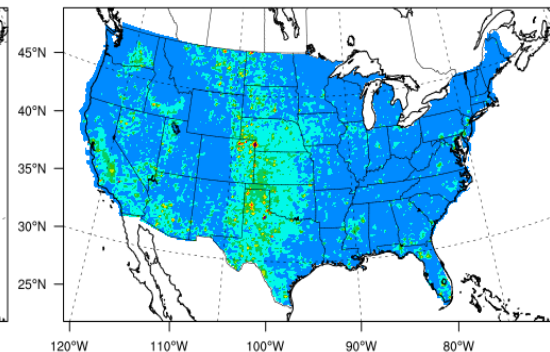
Prior Emissions

Obs mean: 0.09



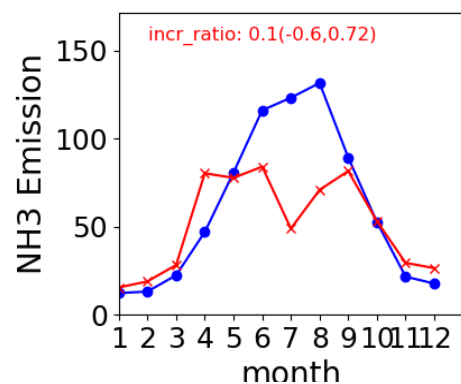
Posterior Emissions

Sim mean: 0.18

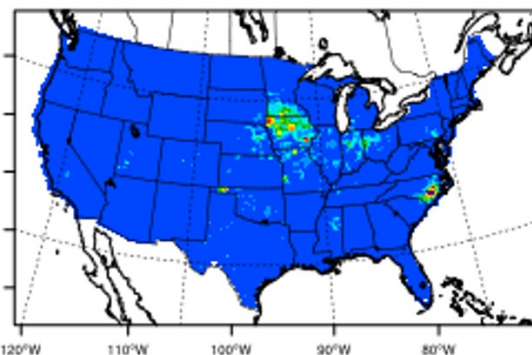


Sector-level NH₃ Emissions Comparison (2019)

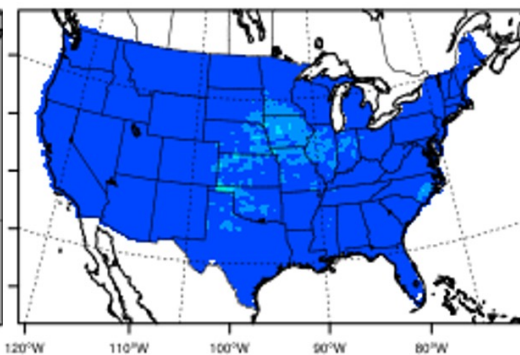
Swine



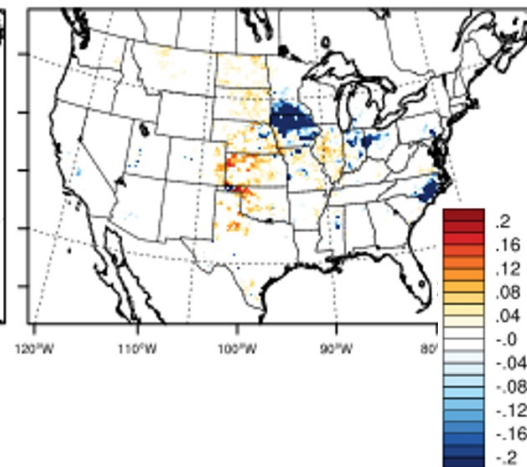
Bottom-up



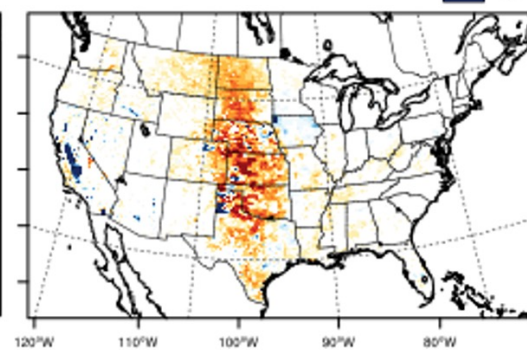
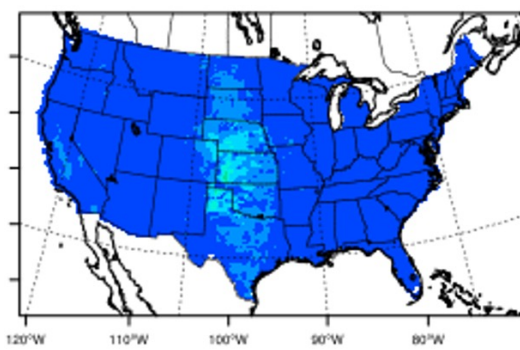
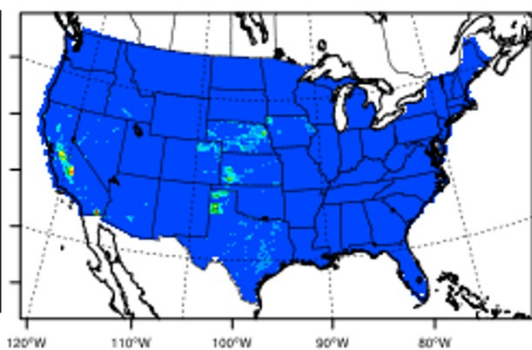
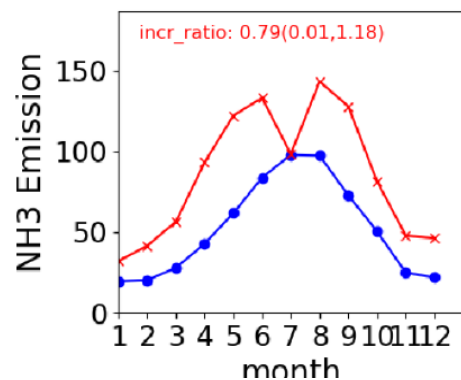
Top-down



Diff (TD-BD)



Beef Cattle



DeepMMF Inverse Modeling Applications

- Improving the overall emissions spatial distributions of pollutants
- Compared to other top-down methods, the DeepMMF can efficiently generate rapidly refreshed, physically-informed top-down emissions within a few hours as soon as observations are ready.
- Can rapidly refresh the emission inputs to enhance the accuracy of daily AQ forecasting simulations.
- With TEMPO satellites hourly products, the DeepMMF can be applied to those stationary high emitters (e.g., Power Plants, Landfill, O&G Refinery, Confined Animal Feeding Operators, Residential Heating,,,) to quantitatively constraint NO_2 and other pollutants
 - Enabling us to develop the hourly resolution temporal profiles for those hotspot emission sources

Questions?

ACKNOWLEDGEMENT

- NASA ACMAP (Atmospheric Composition Modeling and Analysis Program)
- NOAA Atmospheric Chemistry, Carbon Cycle and Climate (AC4)
- Health Effects Institute (HEI) Energy Grant
- Texas Coalition-EDGE
- National Institute for Environmental Research, Korea Ministry of Environment
- USEPA Emission Modeling System Development Contract
- Environment and Climate Change Canada (ECCC)
- Collaborators: Jia Xing from University of Tennessee, Mark Shepard of ECCC, Jesse Bash, Enrico Dammers of TNO